

```
!pip install ultralytics
```

```
Requirement already satisfied: matplotlib>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.5.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0.92)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.11.0+cu128)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.26.0+cu128)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (13.610.43)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.20-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (1.0.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (4.22.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (1.3.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (23.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.5.0->ultralytics) (2.8.2)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.2.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2024.2.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.12.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.5.0)
Requirement already satisfied: setuptools>=82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.0.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.12.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.2)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2024.2.0)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.1)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.7.3.90)
Requirement already satisfied: nvidia-cusparselt-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.93)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.5.0->ultralytics) (1.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.8.0->ultralytics) (2.1.5)
Downloading ultralytics-8.4.84-py3-none-any.whl (1.3 MB)
  1.3/1.3 MB 37.1 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.20-py3-none-any.whl (28 kB)
Installing collected packages: ultralytics-thop, ultralytics
Successfully installed ultralytics-8.4.84 ultralytics-thop-2.0.20
```

```
from ultralytics import YOLO
```

Creating new Ultralytics Settings v0.0.6 file 
 View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
 Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com>

```
model = YOLO("yolo11n.pt")
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt> to 'yolo11n.pt': 100% ————— 5.4MB

```
!ls /content
```

```
Pothole.v1-raw.yolov11.zip runs sample_data yolo11n.pt
```

```
!unzip -q /content/Pothole.v1-raw.yolov11.zip -d /content/
```

```
!ls /content
```

```
data.yaml      README.dataset.txt  runs      test   valid
Pothole.v1-raw.yolov11.zip  README roboflow.txt  sample_data  train  yolo11n.pt
```

```
!cat /content/data.yaml
```

```
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 1
names: ['pothole']

roboflow:
  workspace: brad-dwyer
  project: pothole-voxr1
  version: 1
  license: ODbL v1.0
  url: https://universe.roboflow.com/brad-dwyer/pothole-voxr1/dataset/1
```

```
%%writefile /content/data.yaml
```

```
train: /content/train/images
val: /content/valid/images
test: /content/test/images
```

```
nc: 1
```

```
names:
  0: pothole
```

```
Overwriting /content/data.yaml
```

```
!ls /content/train/images | head
!ls /content/train/labels | head
```

```
img-100_jpg.rf.233751cac3399aa80b5e58a2afbb753a.jpg
img-101_jpg.rf.f213d98946c4f8858a5bf3200c89e39c.jpg
img-102_jpg.rf.16c56ff8f761aec963d9b57aa1b2d3fd.jpg
img-103_jpg.rf.09abf1a8e3477718e143255af3534d29.jpg
img-108_jpg.rf.a35e86abc558a98f252bfc10e49fd6d9.jpg
img-109_jpg.rf.add373710d0d425bd9340c33878e7d42.jpg
img-10_jpg.rf.eaceaaa6f119cf6eb49edc1bc92efaa2.jpg
img-110_jpg.rf.0390f81a2c5b2ad5676a8eb5b043d743.jpg
img-111_jpg.rf.d7e58630e249c45d8c1d564d847dc236.jpg
img-113_jpg.rf.31b41591a5e0c7b0aa02e133e79e1e2a.jpg
img-100_jpg.rf.233751cac3399aa80b5e58a2afbb753a.txt
img-101_jpg.rf.f213d98946c4f8858a5bf3200c89e39c.txt
img-102_jpg.rf.16c56ff8f761aec963d9b57aa1b2d3fd.txt
img-103_jpg.rf.09abf1a8e3477718e143255af3534d29.txt
img-108_jpg.rf.a35e86abc558a98f252bfc10e49fd6d9.txt
img-109_jpg.rf.add373710d0d425bd9340c33878e7d42.txt
img-10_jpg.rf.eaceaaa6f119cf6eb49edc1bc92efaa2.txt
img-110_jpg.rf.0390f81a2c5b2ad5676a8eb5b043d743.txt
img-111_jpg.rf.d7e58630e249c45d8c1d564d847dc236.txt
img-113_jpg.rf.31b41591a5e0c7b0aa02e133e79e1e2a.txt
```

```
from ultralytics import YOLO
```

```
model = YOLO("yolo11n.pt")
```

```
results = model.train(
    data="/content/data.yaml",
    epochs=50,
    imgsz=640,
    batch=16,
    device=0
)
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
43/50	2.92G	1.028	0.8082	1.102	3	640: 100%	30/30 3.6it/s 8.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.767	0.658	0.747	5/5 4.1it/s 1.2s
						0.457	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
44/50	2.92G	1.081	0.8126	1.192	1	640: 100%	30/30 3.7it/s 8.2s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.747	0.689	0.767	5/5 4.4it/s 1.1s
						0.474	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
45/50	2.92G	0.9688	0.7428	1.061	1	640: 100%	30/30 4.3it/s 7.0s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.783	0.7	0.784	5/5 4.9it/s 1.0s
						0.485	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
46/50	2.92G	0.9906	0.7138	1.068	4	640: 100%	30/30 3.6it/s 8.3s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.779	0.664	0.77	5/5 4.4it/s 1.1s
						0.473	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
47/50	2.92G	0.9766	0.7207	1.066	1	640: 100%	30/30 3.6it/s 8.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.763	0.7	0.789	5/5 4.9it/s 1.0s
						0.476	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
48/50	2.92G	0.955	0.7079	1.06	3	640: 100%	30/30 4.2it/s 7.1s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.798	0.676	0.787	5/5 3.4it/s 1.5s
						0.488	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
49/50	2.92G	0.9531	0.7152	1.055	4	640: 100%	30/30 4.2it/s 7.2s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.815	0.656	0.778	5/5 4.2it/s 1.2s
						0.485	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
50/50	2.92G	0.9036	0.6672	1.046	1	640: 100%	30/30 3.6it/s 8.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	133	330	0.784	0.691	0.782	5/5 5.2it/s 1.0s
						0.489	

50 epochs completed in 0.147 hours.
Optimizer stripped from /content/runs/detect/train-2/weights/last.pt, 5.5MB
Optimizer stripped from /content/runs/detect/train-2/weights/best.pt, 5.5MB

Validating /content/runs/detect/train-2/weights/best.pt...
Ultralytics 8.4.84 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO11n summary (fused): 101 layers, 2,582,347 parameters, 0 gradients, 6.3 GFLOPs

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	5/5 2.1it/s 2.4s
all	133	330	0.782	0.691	0.782	0.49	

Speed: 0.2ms preprocess, 3.8ms inference, 0.0ms loss, 3.9ms postprocess per image
Results saved to /content/runs/detect/train-2

!ls /content

data.yaml	README.dataset.txt	sample_data	valid
detect_road_1.webp	README.roboflow.txt	test	yolo11n.pt
Pothole.v1-raw.yolov11.zip	runs	train	yolo26n.pt

```
from ultralytics import YOLO

# Load your trained model
model = YOLO("/content/runs/detect/train-2/weights/best.pt")

# Predict on the uploaded image
results = model.predict(
    source="/content/detect_road_1.webp",
    conf=0.25,
    save=True,
    show_labels=True,
    show_conf=True
)
```

image 1/1 /content/detect_road_1.webp: 448x640 3 potholes, 52.7ms
Speed: 2.2ms preprocess, 52.7ms inference, 1.3ms postprocess per image at shape (1, 3, 448, 640)
Results saved to /content/runs/detect/predict

```
from PIL import Image
import matplotlib.pyplot as plt
```

```
img = Image.open("/content/runs/detect/predict/detect_road_1.webp")

plt.figure(figsize=(12,8))
plt.imshow(img)
plt.axis("off")
plt.show()
```

requirements: Ultralytics requirement ['pi-heif'] not found, attempting AutoUpdate...
 Using Python 3.12.13 environment at: /usr
 Resolved 2 packages in 290ms
 Prepared 1 package in 141ms
 Installed 1 package in 5ms
 + pi-heif==1.4.0

requirements: AutoUpdate success  0.9s

WARNING  **requirements:** Restart runtime or rerun command for updates to take effect

```
-----
FileNotFoundError                                Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/ultralalytics/utils/patches.py in image_open(filename, *args, **kwargs)
    76     try:
--> 77         return _image_open(filename, *args, **kwargs)
    78     except Exception:
```

----- 3 frames -----
FileNotFoundError: [Errno 2] No such file or directory: '/content/runs/detect/predict/detect_road_1.webp'

During handling of the above exception, another exception occurred:

```
FileNotFoundError                                Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/PIL/Image.py in open(fp, mode, formats)
    3511     if is_path(fp):
    3512         filename = os.fspath(fp)
-> 3513         fp = builtins.open(filename, "rb")
    3514         exclusive_fp = True
    3515     else:
```

FileNotFoundError: [Errno 2] No such file or directory: '/content/runs/detect/predict/detect_road_1.webp'

Next steps: [Explain error](#)

```
!find /content/runs/detect -type f
```

```
/content/runs/detect/predict/detect_road_1.jpg
/content/runs/detect/train/args.yaml
/content/runs/detect/train-2/val_batch1_pred.jpg
/content/runs/detect/train-2/train_batch1201.jpg
/content/runs/detect/train-2/args.yaml
/content/runs/detect/train-2/results.png
/content/runs/detect/train-2/train_batch1200.jpg
/content/runs/detect/train-2/BoxR_curve.png
/content/runs/detect/train-2/weights/best.pt
/content/runs/detect/train-2/weights/last.pt
/content/runs/detect/train-2/train_batch2.jpg
/content/runs/detect/train-2/BoxF1_curve.png
/content/runs/detect/train-2/labels.jpg
/content/runs/detect/train-2/val_batch0_labels.jpg
/content/runs/detect/train-2/val_batch1_labels.jpg
/content/runs/detect/train-2/val_batch0_pred.jpg
/content/runs/detect/train-2/confusion_matrix_normalized.png
/content/runs/detect/train-2/BoxP_curve.png
/content/runs/detect/train-2/val_batch2_pred.jpg
/content/runs/detect/train-2/train_batch1202.jpg
/content/runs/detect/train-2/val_batch2_labels.jpg
/content/runs/detect/train-2/confusion_matrix.png
/content/runs/detect/train-2/results.csv
/content/runs/detect/train-2/train_batch0.jpg
/content/runs/detect/train-2/train_batch1.jpg
/content/runs/detect/train-2/BoxPR_curve.png
```

```
!ls -R /content/runs/detect
```

```
/content/runs/detect:
predict train train-2
```

```
/content/runs/detect/predict:
detect_road_1.jpg
```

```

/content/runs/detect/train:
args.yaml  weights

/content/runs/detect/train/weights:

/content/runs/detect/train-2:
args.yaml          results.csv          val_batch0_labels.jpg
BoxF1_curve.png   results.png          val_batch0_pred.jpg
BoxP_curve.png    train_batch0.jpg     val_batch1_labels.jpg
BoxPR_curve.png   train_batch1200.jpg  val_batch1_pred.jpg
BoxR_curve.png    train_batch1201.jpg  val_batch2_labels.jpg
confusion_matrix_normalized.png train_batch1202.jpg val_batch2_pred.jpg
confusion_matrix.png train_batch1.jpg     weights
labels.jpg        train_batch2.jpg

```

/content/runs/detect/train-2/weights:
best.pt last.pt

Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.

```

import glob
from PIL import Image
import matplotlib.pyplot as plt

files = glob.glob("/content/runs/detect/predict/*")

print(files)

img = Image.open(files[0])

plt.figure(figsize=(12,8))
plt.imshow(img)
plt.axis("off")
plt.show()

```

['/content/runs/detect/predict/detect_road_1.jpg']



```
!ls -R /content/runs/detect
```

```
/content/runs/detect:
predict train train-2

/content/runs/detect/predict:
detect_road_1.jpg

/content/runs/detect/train:
args.yaml weights

/content/runs/detect/train/weights:

/content/runs/detect/train-2:
args.yaml          results.csv          val_batch0_labels.jpg
BoxF1_curve.png    results.png          val_batch0_pred.jpg
BoxP_curve.png     train_batch0.jpg     val_batch1_labels.jpg
BoxPR_curve.png    train_batch1200.jpg  val_batch1_pred.jpg
BoxR_curve.png     train_batch1201.jpg  val_batch2_labels.jpg
confusion_matrix_normalized.png train_batch1202.jpg  val_batch2_pred.jpg
confusion_matrix.png train_batch1.jpg     weights
labels.jpg         train_batch2.jpg
```

```
/content/runs/detect/train-2/weights:
best.pt last.pt
```

```
!git config --global user.email "ranjotk.bt.23@nitj.ac.in"
!git config --global user.name "ranjot-kaur123"
```

Start coding or [generate](#) with AI.